

Enriching continuous Lagrange finite element approximation spaces using neural networks

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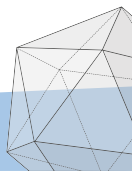
Joint work with:

H. Barucq, F. Faucher, N. Victorion and V. Michel-Dansac.

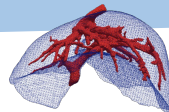
PhD student seminar

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Scientific context

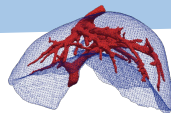


Context: Create real-time digital twins of an organ (e.g. liver).

Objective: Develop an hybrid finite element / neural network method.

accurate quick + parameterized

Scientific context



Context: Create real-time digital twins of an organ (e.g. liver).

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Parametric Poisson problem (with homogeneous Dirichlet BC):

For one or several $\mu \in \mathcal{M}$, find $u : \Omega \rightarrow \mathbb{R}$ such that

$$\begin{cases} -\Delta u(\mathbf{x}; \mu) = f(\mathbf{x}; \mu), & (\mathbf{x}, \mu) \in \Omega \times \mathcal{M}, \\ u(\mathbf{x}; \mu) = 0, & (\mathbf{x}, \mu) \in \partial\Omega \times \mathcal{M}, \end{cases} \quad (\mathcal{P})$$

with $\Omega \subset \mathbb{R}^d$ a domain (d the spatial dimension).

Classical approaches

Physics-Informed Neural Networks

Standard PINNs: Find the optimal weights θ^* , such that

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \left(\omega_r J_r(\theta) + \omega_b J_b(\theta) \right), \quad (\mathcal{P}_\theta)$$

with

residual loss

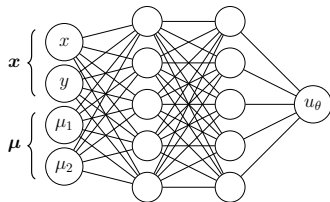
$$J_r(\theta) = \int_{\mathcal{M}} \int_{\Omega} \left| \Delta u_\theta(\mathbf{x}, \boldsymbol{\mu}) + f(\mathbf{x}, \boldsymbol{\mu}) \right|^2 d\mathbf{x} d\boldsymbol{\mu},$$

boundary loss

$$J_b(\theta) = \int_{\mathcal{M}} \int_{\partial\Omega} \left| u_\theta(\mathbf{x}, \boldsymbol{\mu}) \right|^2 d\mathbf{x} d\boldsymbol{\mu},$$

with ω_r and ω_b are some weights.

u_θ is a neural network, e.g. fully-connected NN.
(see example in 2D with 2 parameters)



Monte-Carlo method: Discretize the cost functions by random process.

Finite Element Method¹

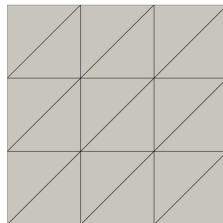
Variational Problem:

$$\text{Find } u_h \in V_h^0 \text{ such that, } \forall v_h \in V_h^0, a(u_h, v_h) = l(v_h), \quad (\mathcal{P}_h)$$

with h the characteristic mesh size, a and l the bilinear and linear forms given by

$$a(u_h, v_h) = \int_{\Omega} \nabla u_h \cdot \nabla v_h, \quad l(v_h) = \int_{\Omega} f v_h,$$

and V_h^0 a continuous, piecewise polynomial space of degree k .



Cartesian mesh

¹[Ern and Guermond, 2004]

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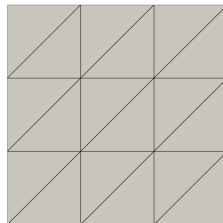
Linear system: Let $(\phi_1, \dots, \phi_{N_{\text{dofs}}})$ a basis of V_h^0 .

Find $U \in \mathbb{R}^{N_{\text{dofs}}}$ such that

$$AU = b,$$

$$A = (a(\phi_i, \phi_j))_{1 \leq i, j \leq N_{\text{dofs}}} \in \mathbb{R}^{N_{\text{dofs}} \times N_{\text{dofs}}},$$

$$b = (l(\phi_j))_{1 \leq j \leq N_{\text{dofs}}} \in \mathbb{R}^{N_{\text{dofs}}}.$$



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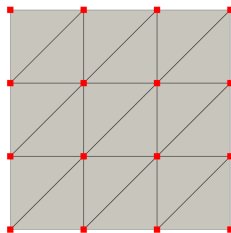
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Degrees of freedom ($k = 1$)

¹[Ern and Guermond, 2004]

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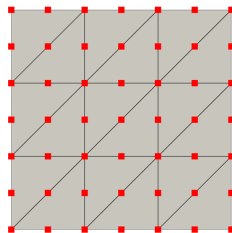
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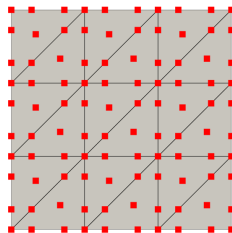
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Degrees of freedom ($k = 3$)

¹[Ern and Guermond, 2004]

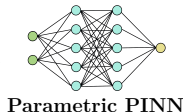
Enriched finite element method using PINNs

Pipeline of the Enriched FEM

OFFLINE : PINN training

Inputs

$\mathbf{x} \in \Omega$: space coordinates
 $\boldsymbol{\mu} \in \mathcal{M}$: parameter



Output

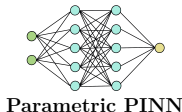
$u_{\theta}(\mathbf{x}, \boldsymbol{\mu})$: prediction of
 the PDE solution $u(\mathbf{x}, \boldsymbol{\mu})$

Pipeline of the Enriched FEM

OFFLINE : PINN training

Inputs

$\mathbf{x} \in \Omega$: space coordinates
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Output

$u_{\theta}(\mathbf{x}, \boldsymbol{\mu})$: prediction of
 the PDE solution $u(\mathbf{x}, \boldsymbol{\mu})$

ONLINE (fixed $\boldsymbol{\mu}$) : PINN evaluation + Enriched FEM resolution

PINN evaluation

$u_{\theta}(\cdot, \boldsymbol{\mu})$: prediction for
 the given parameter $\boldsymbol{\mu}$



FEM solver
 enriched by u_{θ}



Output

enriched FEM solution
 (depending on the mesh size h)

Complete ONLINE process : quick + accurate

Additive approach

Variational Problem: Let $u_\theta \in H^{k+1}(\Omega) \cap H_0^1(\Omega)$ be a PINN prior.

$$\text{Find } c_h^+ \in V_h^0 \text{ such that, } \forall v_h \in V_h^0, a(c_h^+, v_h) = l(v_h) - a(u_\theta, v_h), \quad (\mathcal{P}_h^+)$$

with the **enriched trial space** V_h^+ defined by

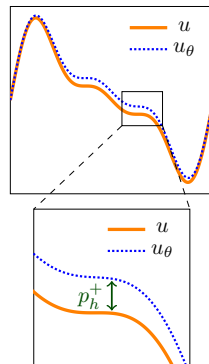
$$V_h^+ = \{u_h^+ = u_\theta + c_h^+, \quad c_h^+ \in V_h^0\}.$$

General Dirichlet BC: If $u = g$ on $\partial\Omega$, then

$$c_h^+ = g - u_\theta \quad \text{on } \partial\Omega,$$

with u_θ the PINN prior.

We expect that the modified problem will give the same results as the standard one on coarser meshes.



Convergence analysis

Theorem 1: Convergence analysis of the standard FEM [Ern and Guermond, 2004]

We denote $u_h \in V_h$ the solution of $(\mathcal{P}_h)$ with V_h the standard trial space. Then,

$$\|u - u_h\|_{L^2} \leqslant C_{L^2} h^{k+1} |u|_{H^{k+1}}.$$

Convergence analysis

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$$\|u - u_h\|_{L^2} \leqslant C_{L^2} h^{k+1} |u|_{H^{k+1}}.$$

Theorem 2: Convergence analysis of the enriched FEM [Lecourtier et al., 2025]

We denote $u_h^+ \in V_h^+$ the solution of $(\mathcal{P}_h^+)$ with V_h^+ the enriched trial space. Then,

$$\|u - u_h^+\|_{L^2} \leqslant \frac{|u - u_\theta|_{H^{k+1}}}{|u|_{H^{k+1}}} (C_{L^2} h^{k+1} |u|_{H^{k+1}}).$$

$$\frac{|u - u_\theta|_{H^{k+1}}}{|u|_{H^{k+1}}} < 1 \Rightarrow \text{Gains of the additive approach.}$$

Numerical results

2D Poisson problem on Square - Dirichlet BC

Problem considered

Problem statement: Consider the Poisson problem in 2D with Dirichlet BC:

$$\begin{cases} -\Delta u = f, & \text{in } \Omega \times \mathcal{M}, \\ u = 0, & \text{on } \partial\Omega \times \mathcal{M}, \end{cases}$$

with $\Omega = [-0.5\pi, 0.5\pi]^2$ and $\mathcal{M} = [-0.5, 0.5]^2$ ($p = 2$ parameters).

Analytical solution: $\mu = (\mu_1, \mu_2) \in \mathcal{M}$

$$u(\mathbf{x}, \mu) = \exp\left(-\frac{(x - \mu_1)^2 + (y - \mu_2)^2}{2}\right) \sin(2x) \sin(2y).$$

PINN training:

MLP 5 layers (sine - 40, 60, 60, 60, 40); LBFGs optimizer (5000 epochs - 6000 col. pts).

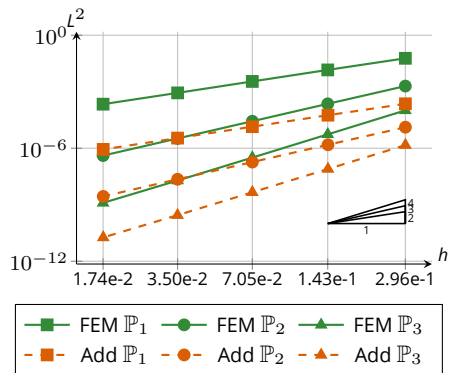
Imposing the Dirichlet BC exactly in the PINN with the levelset φ defined by

$$\varphi(\mathbf{x}) = (x + 0.5\pi)(x - 0.5\pi)(y + 0.5\pi)(y - 0.5\pi).$$

Online phase - 1 parameters instance

$$\mu^{(1)} = (0.05, 0.22)$$

Error estimates in L^2 norm:



Parametric framework

Considering the set of $n_p = 50$ parameters instances

$$\mathcal{S} = \left\{ \boldsymbol{\mu}^{(1)}, \dots, \boldsymbol{\mu}^{(n_p)} \right\}.$$

Gains achieved: $\|u - u_h\|_{L^2} / \|u - u_h^+\|_{L^2}$

Gains in L^2 rel error of our method w.r.t. FEM			
k	min	max	mean
1	134.32	377.36	269.39
2	67.02	164.65	134.85
3	39.52	72.65	61.55

Cartesian mesh: N^2 nodes with $N = 20$.

Numerical results

3D Poisson problem on Cube - Dirichlet BC

Problem considered

Problem statement: Consider the Poisson problem in 3D with Dirichlet BC:

$$\begin{cases} -\Delta u = f, & \text{in } \Omega \times \mathcal{M}, \\ u = 0, & \text{on } \partial\Omega \times \mathcal{M}, \end{cases}$$

with $\Omega = [-0.5\pi, 0.5\pi]^3$ and $\mathcal{M} = [-0.5, 0.5]^3$ ($p = 3$ parameters).

Analytical solution: $\mu = (\mu_1, \mu_2, \mu_3) \in \mathcal{M}$

$$u(\mathbf{x}, \mu) = \exp\left(-\frac{(x - \mu_1)^2 + (y - \mu_2)^2 + (z - \mu_3)^2}{2}\right) \sin(2x) \sin(2y) \sin(2z).$$

PINN training:

MLP 5 layers (tanh - 40, 60, 60, 60, 40); LBFGs optimizer (5000 epochs - 40000 col. pts).

Imposing the Dirichlet BC exactly in the PINN with the levelset φ defined by

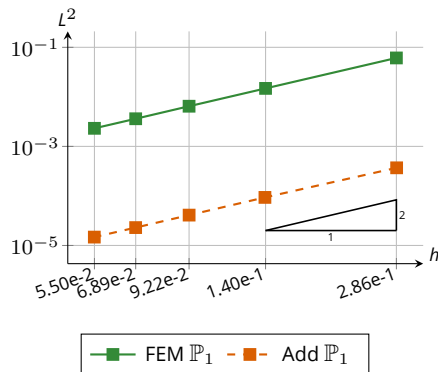
$$\varphi(\mathbf{x}) = (x + 0.5\pi)(x - 0.5\pi)(y + 0.5\pi)(y - 0.5\pi)(z + 0.5\pi)(z - 0.5\pi).$$

Training time: 12 minutes on NVIDIA RTX 2000 (8GB VRAM).

Online phase - 1 parameters instance I

$$\mu^{(1)} = (0.05, 0.22, 0.1)$$

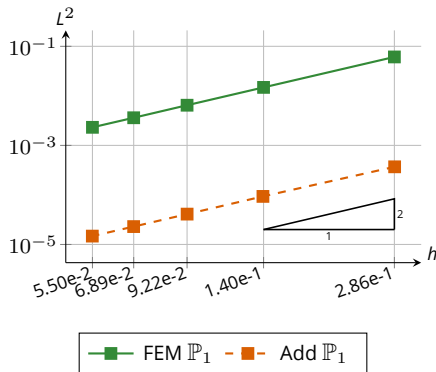
Error estimates in L^2 norm:



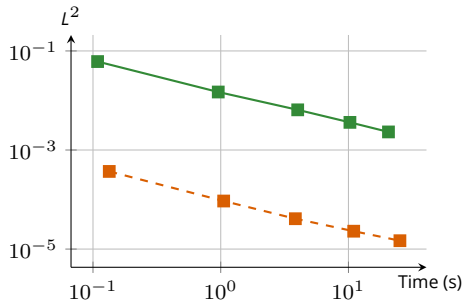
Online phase - 1 parameters instance I

$$\mu^{(1)} = (0.05, 0.22, 0.1)$$

Error estimates in L^2 norm:



L^2 error w.r.t. execution time*:



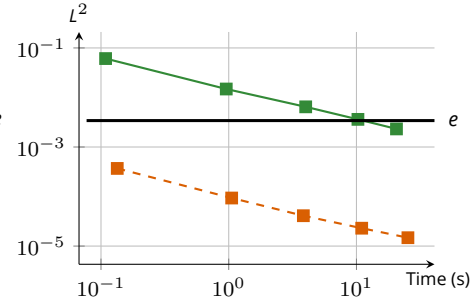
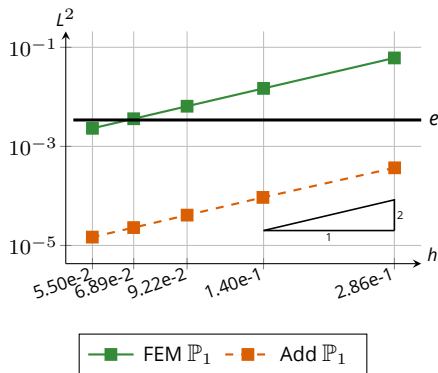
* mesh + assembly + solve
 + prior derivatives evaluation
 + training time

Online phase - 1 parameters instance I

$$\mu^{(1)} = (0.05, 0.22, 0.1)$$

Error estimates in L^2 norm:

L^2 error w.r.t. execution time*:



* mesh + assembly + solve
 + prior derivatives evaluation
 + training time

Online phase - 1 parameters instance II

$$\mu^{(1)} = (0.05, 0.22, 0.1)$$

N_{dofs} and computation time required to reach the same error e :

e	N		N_{dofs}		Computation time*	
	FEM	Add	FEM	Add	FEM	Add
$1 \cdot 10^{-3}$	152	12	$3.51 \cdot 10^6$	$1.73 \cdot 10^3$	$(7.65 \cdot 10^1)$	$5.2 \cdot 10^{-2}$
$1 \cdot 10^{-4}$	484	39	$1.13 \cdot 10^8$	$5.93 \cdot 10^4$	$(2.8 \cdot 10^3)$	$9.26 \cdot 10^{-1}$
$1 \cdot 10^{-5}$	1539	122	$3.65 \cdot 10^9$	$1.82 \cdot 10^6$	$(1.02 \cdot 10^5)$	$(5.26 \cdot 10^1)$

— $\div 1471$ —→

Cartesian mesh: N^3 nodes; $\mathbb{P}_1$ elements.

* The results in brackets are extrapolations.

Numerical results

2D Poisson problem on Square - Dirichlet BC

3D Poisson problem on Cube - Dirichlet BC

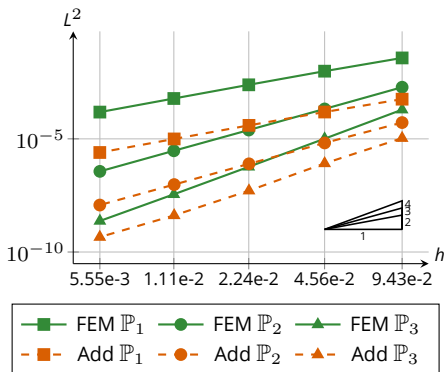
2D Anisotropic Elliptic problem on Square - Dirichlet BC

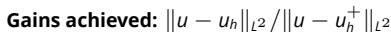
2D Poisson problem on Annulus - Mixed BC

Online phase

Error estimates: 1 parameter instance

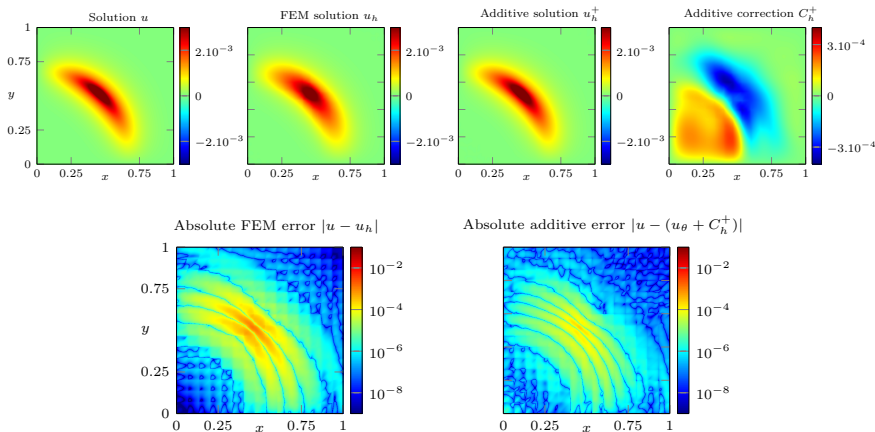
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$$\mathcal{S} = \left\{ \boldsymbol{\mu}^{(1)}, \dots, \boldsymbol{\mu}^{(n_p)} \right\}.$$

Cartesian mesh: N^2 nodes with $N = 20$.

Numerical solutions and errors



$$\mu^{(2)} = (0.46, 0.52, 0.05, 0.12) \quad (k = 2, N = 16)$$

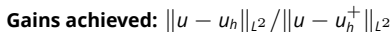
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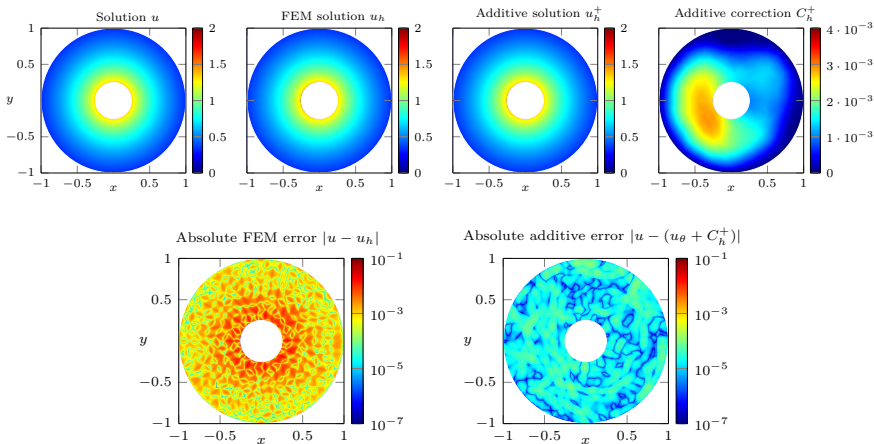
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2D Poisson problem on Annulus - Mixed BC

$$\mu^{(1)} = 2.51$$

$$\mathcal{S} = \left\{ \boldsymbol{\mu}^{(1)}, \dots, \boldsymbol{\mu}^{(n_p)} \right\}.$$

Mesh size: $h = 1.33 \cdot 10^{-1}$

Numerical solutions and errors



$$\mu^{(1)} = 2.51 \quad (k = 1, h = 1.67 \cdot 10^{-1})$$

Extension to non-linear problems

Physics-Informed Neural Network (PINN)

Finite Element Method (FEM)

Enriched finite element method using PINNs

Heated cavity test case¹

Stationary incompressible Navier-Stokes equations (with buoyancy and gravity):

We consider $\Omega = [-1, 1]^2$ a squared domain and $\mathbf{e}_y = (0, 1)$.

Find the velocity $\mathbf{u} = (u_1, u_2)$, the pressure p and the temperature T such that

$$\begin{cases} (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \Delta \mathbf{u} - g(\beta T + 1) \mathbf{e}_y = 0 & \text{in } \Omega & \text{(momentum)} \\ \nabla \cdot \mathbf{u} = 0 & \text{in } \Omega & \text{(incompressibility)} \\ \mathbf{u} \cdot \nabla T - k_f \Delta T = 0 & \text{in } \Omega & \text{(energy)} \\ + \text{suitable BC} & & \end{cases} \quad (\mathcal{P})$$

with $g = 9.81$ the gravity, $\beta = 0.1$ the expansion coefficient, $\nu = 0.01$ the viscosity and $k_f = 0.01$ the thermal conductivity. (Rayleigh number: 156 960)

¹ Non-parametric study.

Heated cavity test case¹

Stationary incompressible Navier-Stokes equations (with buoyancy and gravity):

We consider $\mathbf{x} = (x, y) \in \Omega$ and $\mathbf{e}_y = (0, 1)$.

Find $\mathbf{U} = (\mathbf{u}, p, T) = (u_1, u_2, p, T)$ such that

$$\begin{cases} R_{mom}(\mathbf{U}; \mathbf{x}, \boldsymbol{\mu}) = 0 & \text{in } \Omega & \text{(momentum)} \\ R_{inc}(\mathbf{U}; \mathbf{x}, \boldsymbol{\mu}) = 0 & \text{in } \Omega & \text{(incompressibility)} \\ R_{ener}(\mathbf{U}; \mathbf{x}, \boldsymbol{\mu}) = 0 & \text{in } \Omega & \text{(energy)} \end{cases} \quad (\mathcal{P})$$

with $g = 9.81$ the gravity, $\beta = 0.1$ the expansion coefficient, $\nu = 0.01$ the viscosity and $k_f = 0.01$ the thermal conductivity. (Rayleigh number: 156 960)

Boundary Conditions:

No-slip BC : $\mathbf{u} = 0$ on $\partial\Omega$ **Isothermal BC :** $T = 1$ on the left wall ($x = -1$)
 $T = -1$ on the right wall ($x = 1$)

Adiabatic BC : $\frac{\partial T}{\partial n} = 0$ on the top and bottom walls ($y = \pm 1$, denoted by Γ_{ad})

¹Non-parametric study.

Extension to non-linear problems

Physics-Informed Neural Network (PINN)

Finite Element Method (FEM)

PINN training

Approximate the solution of ($\mathcal{P}$) by a PINN : Find the optimal weights θ^* , such that

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \left(J_{inc}(\theta) + J_{mom}(\theta) + J_{ener}(\theta) + J_{ad}(\theta) \right), \quad (\mathcal{P}_\theta)$$

where the different cost functions¹ are defined by

adiabatic condition

$$J_{ad}(\theta) = \int_{\mathcal{M}} \int_{\Gamma_{ad}} \left| \frac{\partial T_\theta(\mathbf{x}, \mu)}{\partial n} \right|^2 d\mathbf{x} d\mu,$$

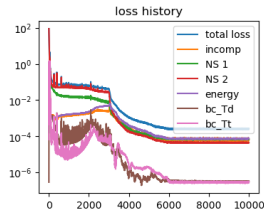
3 residual losses

$$J_\bullet(\theta) = \int_{\mathcal{M}} \int_{\Omega} |R_\bullet(U_\theta(\mathbf{x}, \mu); \mathbf{x}, \mu)|^2 d\mathbf{x} d\mu,$$

with U_θ the non-parametric NN and $\bullet$ the PDE considered (i.e. *inc*, *mom* or *ener*).

Network - MLP		Training (ADAM / LBFGs)			
<i>layers</i>	40, 60, 60, 60, 40	<i>lr</i>	7e-3	N_{col}	40000
σ	sine	<i>nepochs</i>	10000	N_{bc}	30000

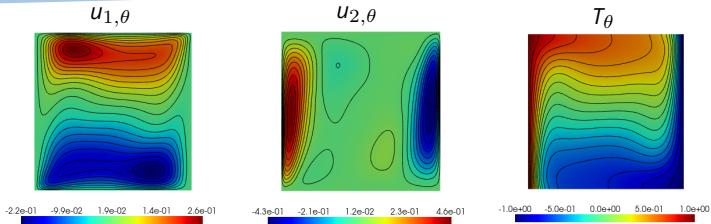
+ some weights to balance the different losses.



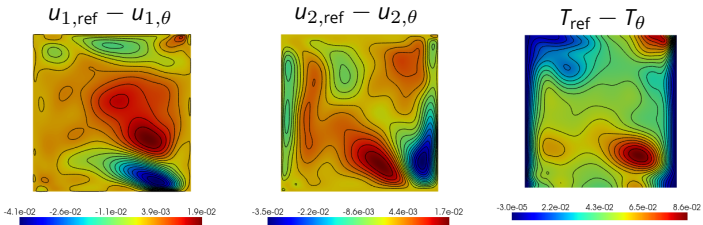
¹ Discretized by a random process using Monte-Carlo method.

Non parametric PINN prediction

Prediction:



Error map¹:



L^2 error:
(relative)

7.60×10^{-2}

5.38×10^{-2}

9.63×10^{-2}

¹Reference solution computed by FEM on an over-refined mesh.

Extension to non-linear problems

Physics-Informed Neural Network (PINN)

Finite Element Method (FEM)

Enriched finite element method using PINNs

Discrete variational formulation

We consider a mixed finite element space $M_h = [v_h^0]^2 \times Q_h \times W_h$ (4 unknowns).

Variational Problem: Find $U_h = (\mathbf{u}_h, p_h, T_h) \in M_h$ s.t., $\forall (\mathbf{v}_h, q_h, w_h) \in M_h^0$,

$$\begin{aligned}
 & \int_{\Omega} (\mathbf{u}_h \cdot \nabla) \mathbf{u}_h \cdot \mathbf{v}_h \, d\mathbf{x} + \mu \int_{\Omega} \nabla \mathbf{u}_h : \nabla \mathbf{v}_h \, d\mathbf{x} \\
 & \quad - \int_{\Omega} p_h \nabla \cdot \mathbf{v}_h \, d\mathbf{x} - g \int_{\Omega} (1 + \beta T_h) \mathbf{e}_y \cdot \mathbf{v}_h \, d\mathbf{x} = 0, \quad (\text{momentum}) \\
 & \int_{\Omega} q_h \nabla \cdot \mathbf{u}_h \, d\mathbf{x} + 10^{-4} \int_{\Omega} q_h p_h \, d\mathbf{x} = 0, \quad (\text{incompressibility + pressure penalization}) \\
 & \int_{\Omega} (\mathbf{u}_h \cdot \nabla T_h) w_h \, d\mathbf{x} + \int_{\Omega} k_f \nabla T_h \cdot \nabla w_h \, d\mathbf{x} = 0, \quad (\text{energy})
 \end{aligned} \tag{\mathcal{P}_h}$$

Discrete variational formulation

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Variational Problem: Find $U_h = (\mathbf{u}_h, p_h, T_h) \in M_h$ s.t., $\forall (\mathbf{v}_h, q_h, w_h) \in M_h^0$,

$$\begin{aligned} \int_{\Omega} (\mathbf{u}_h \cdot \nabla) \mathbf{u}_h \cdot \mathbf{v}_h \, d\mathbf{x} + \mu \int_{\Omega} \nabla \mathbf{u}_h : \nabla \mathbf{v}_h \, d\mathbf{x} \\ - \int_{\Omega} p_h \nabla \cdot \mathbf{v}_h \, d\mathbf{x} - g \int_{\Omega} (1 + \beta T_h) \mathbf{e}_y \cdot \mathbf{v}_h \, d\mathbf{x} = 0, \quad (\text{momentum}) \\ \int_{\Omega} q_h \nabla \cdot \mathbf{u}_h \, d\mathbf{x} + 10^{-4} \int_{\Omega} q_h p_h \, d\mathbf{x} = 0, \quad (\text{incompressibility + pressure penalization}) \\ \int_{\Omega} (\mathbf{u}_h \cdot \nabla T_h) w_h \, d\mathbf{x} + \int_{\Omega} k_f \nabla T_h \cdot \nabla w_h \, d\mathbf{x} = 0, \quad (\text{energy}) \end{aligned} \quad (\mathcal{P}_h)$$

Non-linear terms $\Rightarrow$ Non-linear system to solve.

Newton method

Denoting N_h the dimension of M_h , we want to solve the non linear system:

$$F(\vec{U}) = 0$$

with $F : \mathbb{R}^{N_h} \rightarrow \mathbb{R}^{N_h}$ the non-linear operator associated to $(\mathcal{P}_h)$ and $\vec{U} \in \mathbb{R}^{N_h}$ the unknown vector.

Algorithm 1: Newton algorithm

Initialization step: set $\vec{U}^{(0)} = \vec{U}_0$;

for $n \geq 0$ **do**

 Solve the linear system $F(\vec{U}^{(n)}) + F'(\vec{U}^{(n)})\delta^{(n+1)} = 0$ for $\delta^{(n+1)}$;

 Update $\vec{U}^{(n+1)} = \vec{U}^{(n)} + \delta^{(n+1)}$;

end

Newton method

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end

⚠ The choice of the $\vec{U}_0$ is crucial for the convergence of the Newton method.

Solution: Continuation method where we progressively increase the non-linearity of the problem.

Additive approach

Considering the PINN prior $U_\theta = (\mathbf{u}_\theta, p_\theta, T_\theta)$, we define the **mixed finite element space additively enriched** by the PINN as follows:

$$M_h^+ = \{U_h^+ = U_\theta + C_h^+, \quad C_h^+ \in M_h^0\}$$

with $M_h^0 = [V_h^0]^2 \times Q_h \times W_h^0$, $U_h^+ = (\mathbf{u}_h^+, p_h^+, T_h^+) \in M_h^+$ and $C_h^+ = (\mathbf{c}_{h,u}^+, c_{h,p}^+, c_{h,T}^+)$.

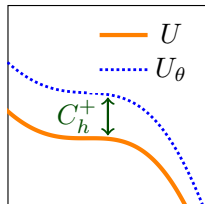
We can then define the three finite element subspaces of M_h^+ as follows:

$$\mathbf{v}_h^+ = \{\mathbf{u}_h^+ = \mathbf{u}_\theta + \mathbf{c}_{h,\mathbf{u}}^+, \mathbf{c}_{h,\mathbf{u}}^+ \in [V_h^0]^2\},$$

$$Q_h^+ = \{p_h^+ = p_\theta + c_{h,p}^+, c_{h,p}^+ \in Q_h\},$$

$$W_h^+ = \{T_h^+ = T_\theta + C_{h,T}^+, C_{h,T}^+ \in W_h^0\},$$

where $\mathbf{C}_{h,u}^+$, $C_{h,p}^+$ and $C_{h,T}^+$ becomes the unknowns.



Newton method - Additive approach

Denoting N_h the dimension of M_h^+ , we want to solve the non-linear system:

$$F_\theta(\vec{C}) = 0$$

with $F_\theta : \mathbb{R}^{N_h} \rightarrow \mathbb{R}^{N_h}$ the non linear operator associated to the weak problem $(\mathcal{P}_h^+)$ and $\vec{C} \in \mathbb{R}^{N_h}$ the **correction vector (unknown)**.

Algorithm 2: Newton algorithm

Initialization step: set $\vec{C}^{(0)} = \mathbf{0}$;

for $n \geq 0$ **do**

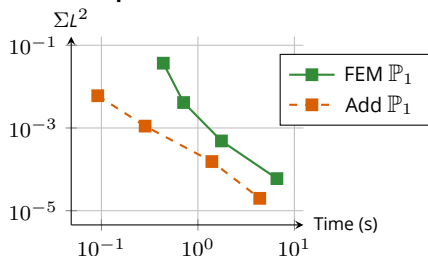
 Solve the linear system $F_\theta(\vec{C}^{(n)}) + F'_\theta(\vec{C}^{(n)})\delta^{(n+1)} = 0$ for $\delta^{(n+1)}$;

 Update $\vec{C}^{(n+1)} = \vec{C}^{(n)} + \delta^{(n+1)}$;

end

Numerical results

N_{dofs} and computation time required to reach the same error e :



e	N_{dofs}		Online time	
	FEM	Add	FEM	Add
$1 \cdot 10^{-3}$	33 204	13 764	1.29	0.31
$1 \cdot 10^{-4}$	150 339	70 303	4.76	1.78
$1 \cdot 10^{-5}$	690 924	339 231	20.34	6.42

Conclusion

TODO

Preprint (linear)



References

- A. Ern and J.-L. Guermond. Theory and Practice of Finite Elements. Springer New York, 2004.
- F. Lecourtier, H. Barucq, M. Duprez, F. Faucher, E. Franck, V. Lleras, V. Michel-Dansac, and N. Victorion. Enriching continuous lagrange finite element approximation spaces using neural networks, 2025.
- N. Sukumar and A. Srivastava. Exact imposition of boundary conditions with distance functions in physics-informed deep neural networks. Comput. Method. Appl. M., 389:114333, 2022. ISSN 0045-7825.
- M. Tancik, P. Srinivasan, and al. Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains. In Advances in Neural Information Processing Systems, volume 33, pages 7537–7547. Curran Associates, Inc., 2020.

Preprint (linear)

